Thermal Analysis

Thermal Analysis and CFD Simulation of a Small Satellite Cooling System

Final report

Master's Degree in Computational Fluid Dynamics

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Contents

1 Final Report **3**

1.1 Introduction . . . 3
1.2 Geometry Definition . . . 4
1.3 Geometry Modeling . . . 4
1.4 Computational Mesh and Strategy . . . 5
1.4.1 Mesh Quality Validation . . . 6
1.5 Physical Models & Boundary Conditions . . . 6
1.5.1 Fluid Domain Physics . . . 7
1.5.2 Solid Domain Physics . . . 7
1.5.3 Space Thermal Environment and Boundary Conditions . . . 7
1.6 Solution Method and Numerical Strategy . . . 7
1.6.1 Solver Formulation and Governing Equations . . . 7
1.6.2 Radiation Modeling and Environmental Boundary Conditions . . . 8
1.6.3 Fluid Movement Modeling . . . 9
1.7 Mesh Independence Study . . . 9
1.8 Results and Discussion . . . 11
1.8.1 Global Conjugate Heat Transfer and Environmental Solar Asymmetry 11
1.8.2 Internal Fluid Dynamics and Thermal Core Behavior . . . 12
1.8.3 External Chassis Temperature and Radiative Dissipation . . . 13
1.9 Conclusion . . . 14

1

Final Report

1.1 Introduction

Small satellite platforms, particularly those conforming to the CubeSat standard, have emerged as cost-effective solutions for a wide range of Low Earth Orbit (LEO) missions. However, their highly compact form factor imposes stringent constraints on thermal management: the limited external surface area must dissipate both internal electronic heat loads and the external solar irradiation absorbed during the orbital cycle, while maintaining all components within strict operational temperature limits.

This work presents a comprehensive multi-physics CFD simulation of an active single-phase liquid cooling loop integrated into a small satellite structure, developed using Simcenter STAR-CCM+. The analysis targets the *sunlit orbital phase* in LEO, representing a thermally demanding operational scenario in which the satellite simultaneously experiences an internal electronic heat generation of $Q_{\text{src}} = 1.0\,\text{W}$ and a directional solar irradiation load $G_{\text{solar}} \approx 1360\,\text{W/m}^2$, of which a fraction $q_{\text{solar}} = \alpha \cdot G_{\text{solar}} = 408\,\text{W/m}^2$ is absorbed by the exposed aluminum face ($\alpha = 0.30$), while all external surfaces continuously reject heat into the deep space thermal sink at $T_{\text{space}} = 3.0\,\text{K}$ via surface radiation.

The three coupled heat transfer mechanisms governing the system are simultaneously resolved: *solid conduction* through the metallic chassis and mounting brackets, *forced convection* within the circulating liquid coolant, and *surface radiation* from the external aluminum panels to the space environment.

The primary objective is to verify that the on-board electronics, modeled as a 1.0 W uniform volumetric heat source, remain within the safe operational temperature range of $-20\,^\circ\text{C}$ to $+60\,^\circ\text{C}$ (253.15–333.15 K) under this combined thermal loading. Secondary objectives include characterizing the fluid velocity distribution within the cooling loop, assessing the convective heat extraction efficiency, and evaluating the asymmetric thermal response of the external chassis to the directional solar input.

1.2 Geometry Definition

The geometry was developed and refined within the STAR-CCM+ 3D-CAD modeler. The satellite structure consists of a simplified aluminum frame, which serves as the primary conduction path for the assembly. Inside this frame, three acrylic plates are located on the right, left, and back faces (leaving three faces made only of aluminum without acrylic). These plates act as thermal insulating material, directing heat preferentially toward the active cooling path.

To simulate the internal thermal load, a centralized internal solid plate was implemented. This component acts as a global thermal representative, mimicking the cumulative heat generation of the on-board electronics (PCBs, batteries, and processing units).

A perimeter cooling loop is routed along the frame and is thermally coupled through four solid mounting brackets. This layout ensures efficient heat transport from the internal heat source to the external radiating surfaces. External appendages such as solar panels are omitted to keep the simulation focused on the cooling loop performance, and the loop is modeled as a single-phase liquid flow driven by a virtual pump. The overall structural concept is consistent with typical CubeSat mechanical layouts.

1.3 Geometry Modeling

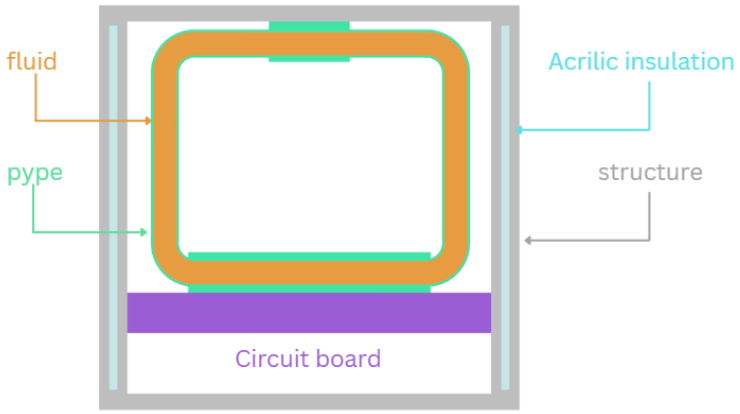


Figure 1.1: CubeSat cross-sectional view in STAR-CCM+, showing (from outside inward) the aluminum structural frame (grey), acrylic insulation plates (cyan), perimeter cooling pipe (orange), internal fluid domain (yellow), and central electronics board (purple) acting as the 1.0 W volumetric heat source.

The cross-section of the CubeSat geometry is illustrated in Fig. 1.1. The model is designed to simulate the thermal path from internal heat-generating components to the external radiating surfaces.

Pipe: The pipe runs along the internal perimeter of the structure, bridging the heat source and the dissipation panels. Given its circular cross-section, the contact region is modeled as a small rectangular parallelepiped to ensure a robust surface-to-surface interface for mesh generation.

Fluid Domain: The internal volume of the cooling pipe, where the liquid coolant facilitates heat transport through forced convection.

Heat Source: A solid block representing the electronic components, modeled as a uniform volumetric heat source. Heat is transferred via conduction to the structure and subsequently dissipated through radiation.

Thermal Insulation: Acrylic plates on the lateral sides ensure thermal flux is preferentially directed through the heat pipe toward the primary radiating panels.

1.4 Computational Mesh and Strategy

The spatial discretization of the computational domain was executed using the Automated Mesh operation within Simcenter STAR-CCM+. A hybrid meshing strategy was selected to provide an optimal balance between numerical accuracy and total cell count, tailoring the grid topology to the specific geometric and physical requirements of each subsystem.

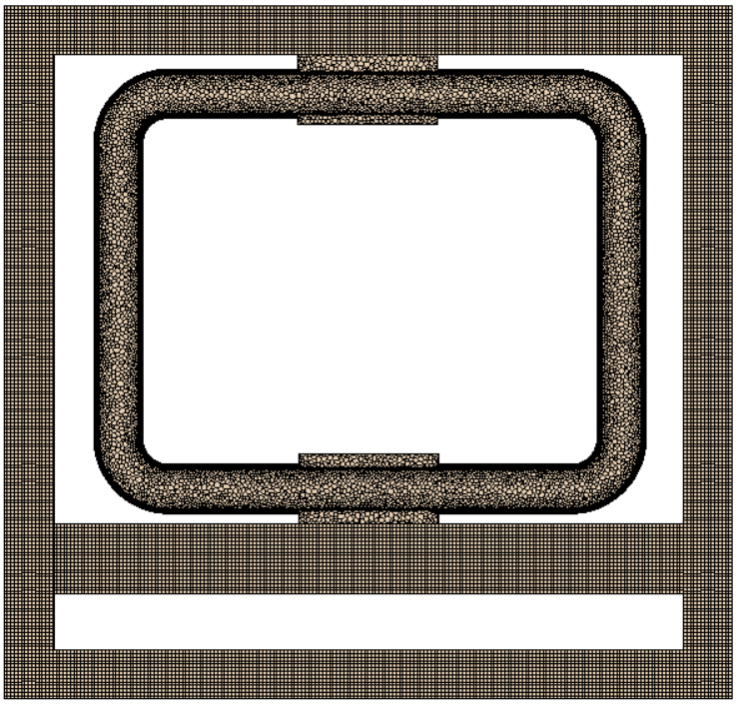


Figure 1.2: Global view of the computational mesh, showing the Trimmed Cell Mesher applied to the structural solids and the Polyhedral/Prism Layer pipeline applied to the perimeter cooling loop.

The core structural components — comprising the aluminum chassis, the electronics heat

source, and the acrylic insulation plates — were discretized using the Trimmed Cell Mesher and Surface Remesher. This Cartesian-based approach is highly effective for orthogonal geometries, with a uniform baseline size of 5×10^{-4} m enforced across these solid domains.

For the fluid-conveying conduits, more advanced meshing pipelines were required. The external pipe features an extremely small wall thickness, accurately captured by the Thin Mesher (five thin layers) supplemented by the Polyhedral Mesher and Surface Remesher. The internal pipe (the sole fluid domain) was discretized using the Prism Layer Mesher to resolve the hydrodynamic and thermal boundary layers at the fluid-solid interface, with the core finalized using the Polyhedral Mesher, Surface Remesher, and Generalized Cylinder Mesher.

1.4.1 Mesh Quality Validation

The *Cell Quality* metric was evaluated throughout the domain to ensure solution reliability.

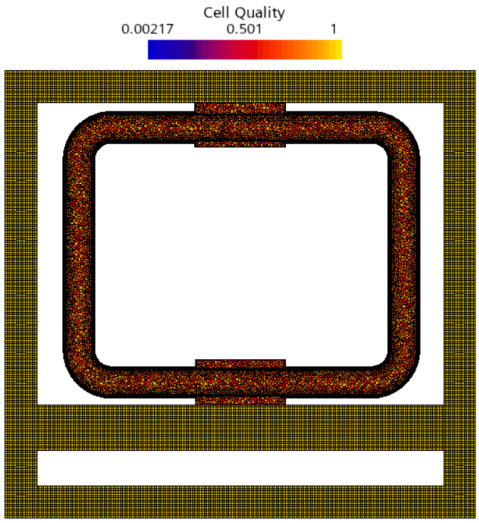


Figure 1.3: Cell Quality distribution across the CubeSat section. The vast majority of cells exhibit a quality index close to 1.0, confirming a well-conditioned grid throughout the domain.

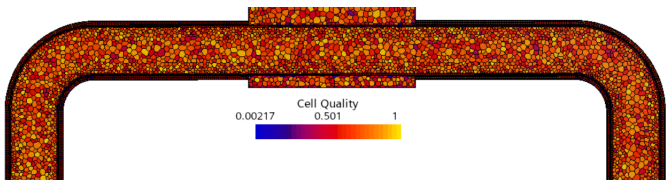


Figure 1.4: Detailed view of the pipe cross-section, showing the structured Prism Layers at the wall boundary (near-zero quality gradient) and the polyhedral core in the interior of the fluid domain.

As shown in Figs. 1.3 and 1.4, the cell quality remains high throughout the domain, particularly at the fluid-solid interfaces. The minimum cell quality value is well above the critical thresholds, ensuring an optimal mesh for solver convergence.

1.5 Physical Models & Boundary Conditions

This section details the mathematical and physical formulations selected within Simcenter STAR-CCM+ to characterize the thermal and fluid dynamics behavior of the CubeSat. The

computational physics is partitioned into distinct continua, isolating the fluid domain from the solid structures while establishing rigorous thermal coupling at their interfaces.

1.5.1 Fluid Domain Physics

The internal cooling loop is governed by the continuous flow of a single-phase liquid refrigerant in steady-state laminar flow. Given the characteristic dimensions of the micro-channels and the low operational flow rates typical of aerospace micro-pumps, the Reynolds number falls well within the laminar regime. The fluid physics continuum includes the following primary models:

- **Liquid:** Constant density and temperature-dependent properties for the specific coolant.
- **Steady:** Time-independent solver suited for continuous operational conditions.
- **Laminar Flow:** Appropriate for low Reynolds number regimes inside the small-caliber tubing.
- **Fluid Energy:** Coupled thermal energy solver to calculate convective heat absorption.

1.5.2 Solid Domain Physics

The solid structures — comprising the external aluminum chassis, the internal mounting brackets, and the electronic board — are governed purely by the steady-state heat conduction equation. The thermal properties of the aluminum components are assumed isotropic, providing an efficient path for conductive heat transfer. A uniform volumetric heat source ($\mathrm{W/m^3}$) is applied to the electronic board domain, calibrated to release a total thermal power of 1.0 W.

1.5.3 Space Thermal Environment and Boundary Conditions

Operating in LEO orbit, the CubeSat is completely isolated from any ambient convective medium. Thermal management relies on conductive paths within the solid structure and subsequent radiative rejection from the outer surfaces. The external space is defined as an ideal thermal sink at $T_{\mathrm{space}} = 3.0\,\mathrm{K}$, with the convective coefficient on all external boundaries set to zero. In addition, the sun is modeled as a constant heat flux impinging on one specific external face, representing the directional solar load during the sunlit orbital phase.

1.6 Solution Method and Numerical Strategy

1.6.1 Solver Formulation and Governing Equations

A steady-state, pressure-based segregated solver is employed, solving sequentially the continuity, momentum, and energy equations with SIMPLE pressure-velocity coupling. Second-order upwind schemes are applied to the convective terms of both momentum and energy equations.

Given that the liquid coolant operates at low velocities ($Ma \ll 0.3$), an incompressible fluid model is fully justified.

1.6.2 Radiation Modeling and Environmental Boundary Conditions

To simulate the radiative heat interaction between the CubeSat's outer shell and the space environment without the computational overhead of full surface-to-surface ray-tracing solvers, a direct mathematical boundary condition was implemented via User Field Functions in STAR-CCM+.

The satellite geometry comprises six external faces. Only one face is assumed to be directly exposed to solar irradiation, representing the sunlit orbital phase in which a single panel faces the Sun. The remaining five faces are considered to be in shadow, receiving no direct solar flux.

For the shadowed panels, the boundary condition models pure radiative emission to deep space:

$$q_{\text{net}} = -\sigma \cdot \varepsilon \cdot \left(T^4 - T_{\text{space}}^4\right) \tag{1.1}$$

implemented in STAR-CCM+ as the following User Field Function:

```
-1.0 * 5.67e-8 * 0.85 * (pow(${Temperature}, 4) - pow(3.0, 4))
```

For the sun-exposed panel, both solar absorption and thermal emission are accounted for simultaneously:

$$q_{\text{net}} = \alpha \cdot G_{\text{solar}} - \sigma \cdot \varepsilon \cdot \left(T^4 - T_{\text{space}}^4\right) \tag{1.2}$$

implemented as:

```
408 - 1.0 * 5.67e-8 * 0.85 * (pow(${Temperature}, 4) - pow(3.0, 4))
```

where $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ is the Stefan-Boltzmann constant, $\varepsilon = 0.85$ is the surface emissivity, T is the local wall temperature computed at each solver iteration, and $T_{\text{space}} = 3.0\,\mathrm{K}$ is the fixed deep space sink temperature. The absorbed solar flux is derived from the LEO solar constant $G_{\text{solar}} \approx 1360\,\mathrm{W/m^2}$ and the solar absorptivity of the surface $\alpha \approx 0.30$:

$$q_{\text{solar}} = 0.30 \times 1360\,\mathrm{W/m^2} = 408\,\mathrm{W/m^2} \tag{1.3}$$

It is important to note that $\varepsilon = 0.85$ and $\alpha = 0.30$ are distinct physical properties: ε governs thermal emission in the infrared spectrum, while α describes absorption in the solar (visible and near-UV) spectrum. For aluminum with surface treatment, these values differ significantly, and both must be specified independently. This formulation dynamically couples the local surface temperature to the net outgoing flux at each solver iteration, driving the external surfaces toward the steady-state thermal equilibrium at which internal and external loads are fully balanced.

A recognized limitation of this approach is that it does not capture radiative exchange between satellite surfaces (inter-surface view factors); this would require a view-factor model in a more detailed follow-up study.

1.6.3 Fluid Movement Modeling

To simulate the micro-pump without modeling the complex rotating geometry of an impeller, a macroscopic strategy was adopted: the pump action is represented as a localized volumetric momentum source term ($\mathrm{N/m^3}$) applied to a targeted fluid control volume. A custom vector User Field Function provides a constant and directionally controlled driving force:

$$\vec{S}_m = \frac{\vec{v}}{|\vec{v}|} \cdot 200.0 \tag{1.4}$$

where $\vec{v}$ is the local fluid velocity vector. The velocity unit vector continuously aligns the source with the local flow trajectory, preventing non-orthogonal pressure gradients and numerical backflow. The scalar multiplier of $200.0\,\mathrm{N/m^3}$ provides the mechanical energy required to overcome friction losses across the 90° bends. This approach completely bypasses transient sliding meshes or Moving Reference Frames, significantly reducing computational overhead.

1.7 Mesh Independence Study

Three computational grids of increasing refinement — designated Coarse, Medium, and Fine — were generated and evaluated under identical boundary conditions to verify that the final solution is free of spatial discretization errors. The monitored metrics are the average electronics board temperature (T_{board}) and the average fluid velocity (v_{avg}), summarized in Table 1.1.

Table 1.1: Grid independence study evaluating the sensitivity of key thermal and fluid metrics to mesh refinement.

Configuration	Total Cell Count	Board T [K]	Fluid V [m/s]
Coarse	1,034,055	303.345	0.1003
Medium (selected)	2,610,128	303.341	0.0992
Fine	5,799,488	303.232	0.1004

The relative variation in average fluid velocity between Medium and Fine configurations is approximately 1.23 %, while the absolute temperature discrepancy is only 0.11 K (relative variation of 0.036 %). The Medium mesh was therefore selected as the production grid, providing a grid-independent solution at reduced computational cost.

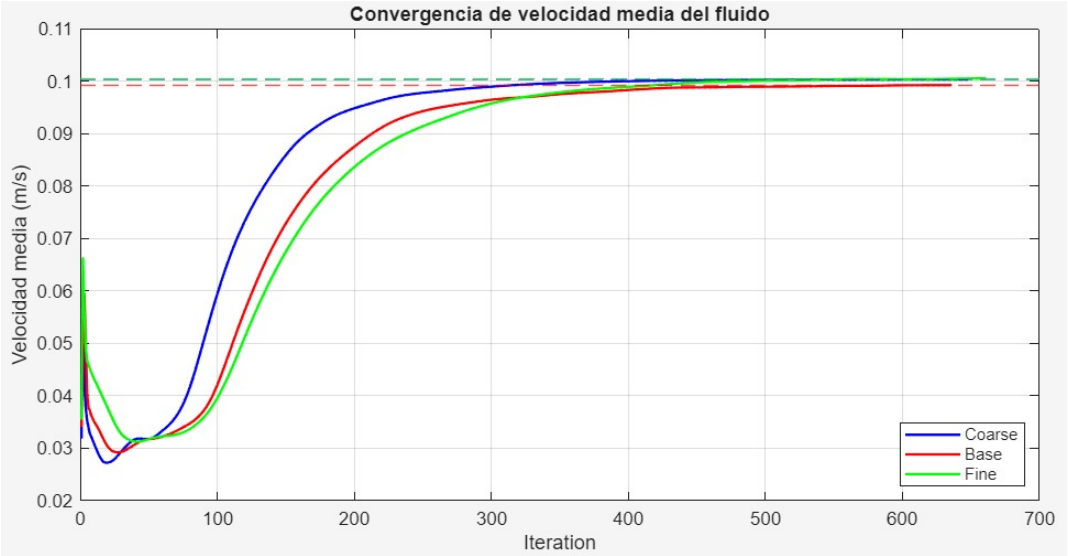


Figure 1.5: Solver convergence history of the average fluid velocity for Coarse, Medium, and Fine mesh configurations. All three grids exhibit an initial transient rise followed by smooth asymptotic stabilization between iterations 450 and 500. The near-identical converged values of the Medium and Fine curves confirm mesh independence of the flow solution.

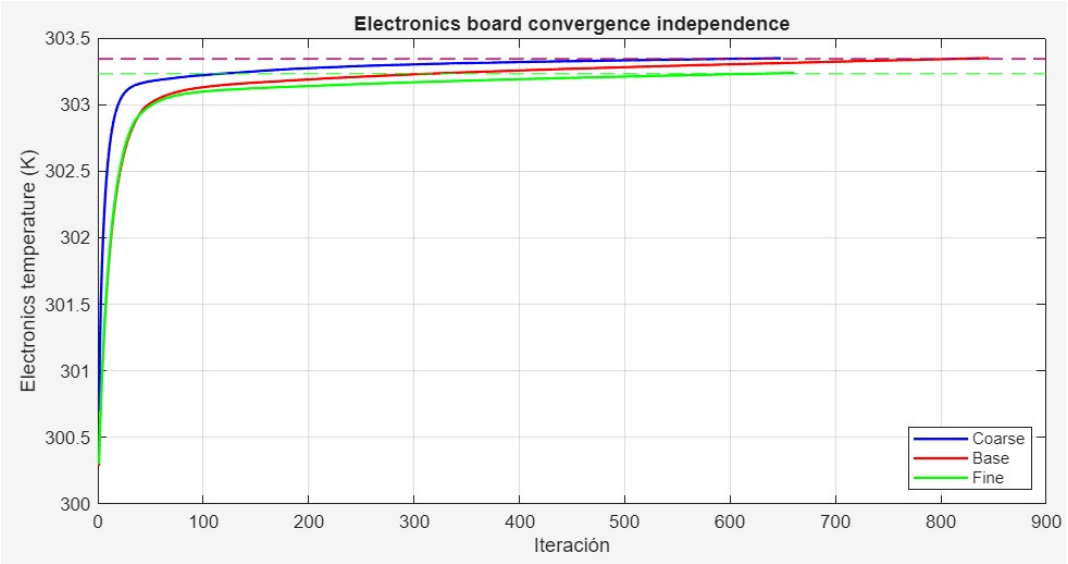


Figure 1.6: Solver convergence history of the electronics board average temperature for Coarse, Medium, and Fine mesh configurations. The solid domain requires 600–800 iterations to fully stabilize, consistent with the larger thermal inertia of the conduction-dominated solid relative to the convective fluid. The overlapping Medium and Fine curves confirm that spatial discretization error is negligible in the selected grid.

1.8 Results and Discussion

This chapter presents the numerical findings obtained from the steady-state conjugate heat transfer simulation. The electronics board temperature (shown in Fig. 1.6) converges asymptotically to a steady-state equilibrium, confirming that the 1.0 W internal heat generation is globally balanced by the combined solar absorption and radiative rejection into deep space. The results are organized around three aspects: the global thermal response, the internal fluid dynamics, and the external radiative dissipation.

1.8.1 Global Conjugate Heat Transfer and Environmental Solar Asymmetry

Before analyzing the localized fluid dynamics of the cooling loop, the macroscopic thermal equilibrium of the entire CubeSat assembly is evaluated through the cross-sectional scalar scene of Fig. 1.7.

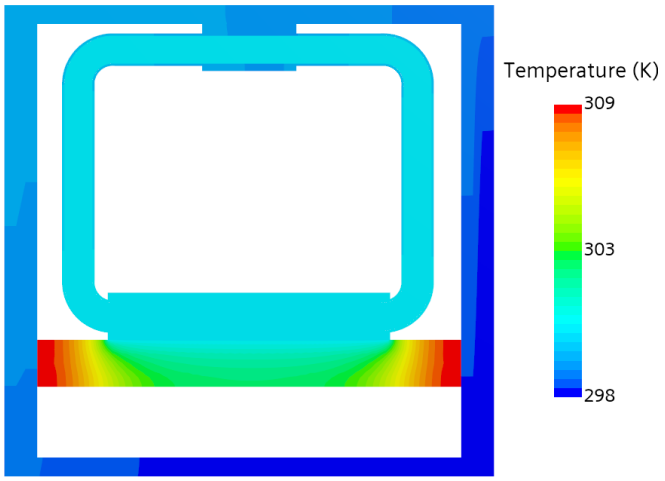


Figure 1.7: Cross-sectional scalar scene of the global temperature distribution at steady state (temperature range: 298–309 K). The electronic board (bottom) acts as the dominant internal heat source, reaching ∼309 K at its lateral extremities and dropping to ∼303–304 K at the interface with the cooling channel. A macroscopic thermal asymmetry is visible on the frame walls: the solar-exposed left panel reaches ∼299.5 K while the shadowed right panel remains at ∼298 K.

Fig. 1.7 illustrates how the external environmental loads and the internal heat generation interact at steady state. Along the exterior boundaries, a macroscopic thermal asymmetry is visible: the left vertical panel of the chassis exhibits a warmer temperature profile ($\approx 299.5\,\mathrm{K}$) compared to the right vertical panel, which remains at a colder baseline of 298.0 K. This non-uniform distribution is the direct numerical result of modeling the directional solar flux hitting the satellite's exterior skin, which penetrates the outer shell and is distributed by conduction through the structural aluminum frame.

Concurrently, Fig. 1.7 reveals the internal conjugate heat transfer mechanisms acting as the primary thermal defense. The electronic board exhibits its highest temperature peaks at its lateral outer extremities, reaching approximately 309.0 K (bright red zones). Crucially, the central portion of the board shows a significant temperature drop toward greener gradients ($\approx$ 303.0–304.0 K) where it interfaces directly with the lower horizontal branch of the cooling channel, proving that the internal fluid loop acts as a dominant thermal sink and establishes a strong, controlled conductive gradient that drives heat inward toward the fluid domain.

1.8.2 Internal Fluid Dynamics and Thermal Core Behavior

The internal velocity field and the resulting fluid temperature distribution are analyzed in a coupled framework, as presented in Figs. 1.8 and 1.9.

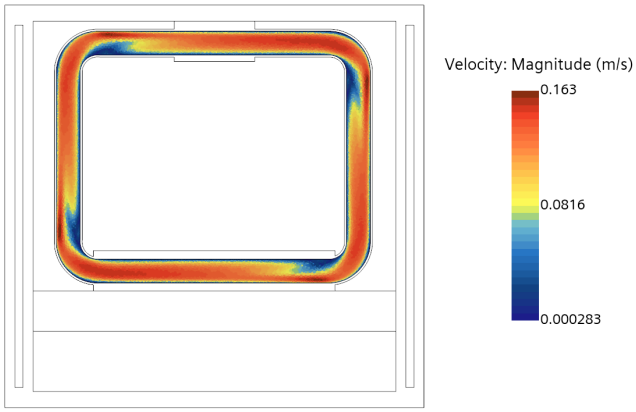


Figure 1.8: Velocity magnitude distribution within the closed liquid cooling loop. Straight segments reach a uniform baseline of 0.12–0.14 m/s (green). Centrifugal forces at the 90° bends push the high-speed core outward, creating local peaks up to 0.163 m/s (red). Flow separation on the inner bend radii produces near-zero stagnation zones of 0.000283 m/s (dark blue), which represent localized convective deficiencies.

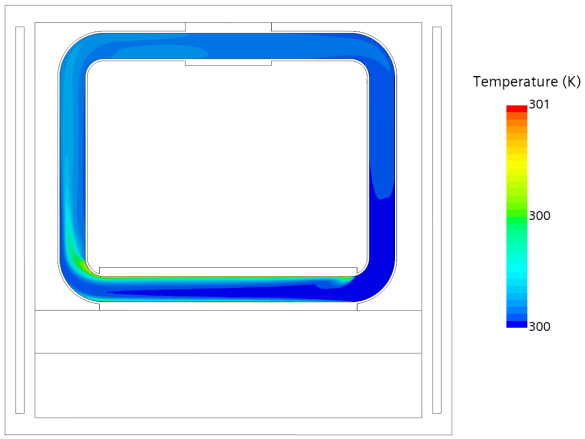


Figure 1.9: Fluid temperature distribution within the cooling loop. The coolant enters the lower branch at ~300 K and absorbs heat from the electronics board, rising to a peak of 301.0 K near the bottom-left bend. The total thermal rise across the full circuit remains below 1.0 K, confirming the absence of coolant thermal saturation and validating the adequacy of the pump flow rate.

The pump momentum source successfully establishes stable fluid recirculation, driving the liquid coolant to a uniform baseline velocity between 0.12 m/s and 0.14 m/s along the straight channels, consistent with aerospace micro-pump operational limits. However, sharp 90° corners introduce critical localized transitions: centrifugal forces push the high-velocity core outward, triggering local acceleration up to 0.163 m/s, while prominent flow separation downstream of the inner radii creates stagnation zones where velocity drops to a minimum of 0.000283 m/s. This hydrodynamic layout directly governs the internal thermal distribution: as the fluid traverses the lower horizontal branch coupled with the 1 W electronics, it absorbs energy and rises to a peak of 301.0 K near the bottom-left bend. The inner stagnation zones restrict

local mixing and cause localized heat accumulation, demonstrating that structural mounting brackets must be placed away from post-corner inner boundaries to avoid thermal bottlenecks.

1.8.3 External Chassis Temperature and Radiative Dissipation

The temperature distribution on the external surfaces of the aluminum chassis is evaluated in Fig. 1.10 to verify the macroscopic thermal equilibrium and heat rejection capabilities under orbital conditions.

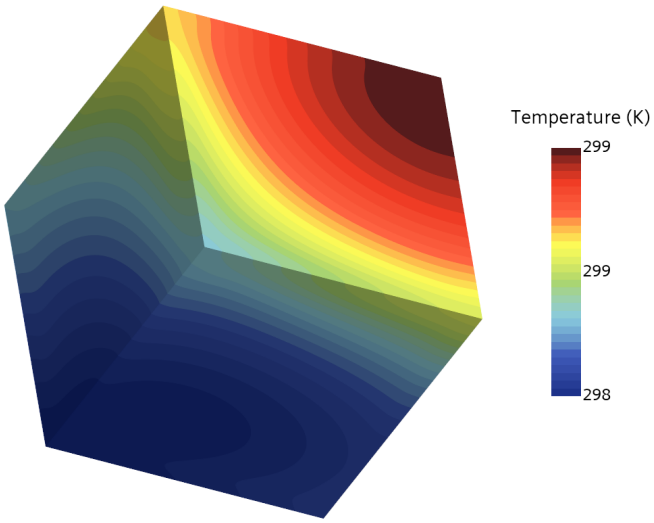


Figure 1.10: Scalar scene of the external chassis temperature distribution at steady state (range: 298–299.5 K). The sun-exposed face (upper right) reaches a peak of ∼299.5 K (dark red/orange) due to the 408 $\mathrm{W/m^2}$ absorbed solar flux, while shadowed panels settle at ∼298.0 K (deep blue). The ≈1.5 K total chassis gradient confirms that aluminum conduction redistributes the directional solar load before the T^4 radiation boundary condition rejects it to the 3.0 K deep space sink.

Fig. 1.10 demonstrates a non-uniform thermal profile across the external skin, directly revealing the impact of the directional solar load. The upper-right surface panel exhibits a prominent hot zone reaching approximately 299.5 K, marking the face directly exposed to the 408 $\mathrm{W/m^2}$ solar influx. Conversely, the adjacent and shadowed panels drop progressively toward a baseline of 298.0 K, establishing a total thermal gradient of roughly 1.5 K across the structural shell. This smooth transition proves that the high-conductivity aluminum frame distributes a portion of the incoming solar load toward the colder faces, while the localized nature of the solar heating forces the mathematical T^4 net radiation boundary condition to emit energy at a higher rate from the warm, sun-lit surface. The external temperature field thus validates the environmental modeling framework, confirming that the chassis successfully balances combined internal heat generation and asymmetric external solar loading before continuously rejecting the net load into deep space.

1.9 Conclusion

This study has evaluated the thermal and fluid-dynamic performance of an active single-phase liquid cooling loop for a small satellite operating under LEO sunlit orbital conditions. Through a verified multi-physics CFD model, the system's capability to maintain all component temperatures within standard safety boundaries under combined internal heat generation (1.0 W) and directional solar loading (408 W/m^2) has been successfully demonstrated.

The key findings of this investigation are:

- **Thermal Validation:** The active cooling loop successfully manages the combined thermal load. The electronic board temperature stabilizes at approximately 303.34 K ($\sim$ 30.19 °C), well within the safe operating range of -20 °C to $+60$ °C, with a comfortable $\sim$30 K margin below the upper limit.
- **Fluid-Dynamic Efficiency:** The localized momentum source sustains a stable average fluid velocity of 0.099 m/s, with local peaks of 0.163 m/s at the bend outer radii. This flow maintains the total coolant temperature rise within 1.0 K across the full loop, preventing thermal saturation of the coolant.
- **Environmental Radiative Balance:** The directional solar irradiation induces a chassis thermal gradient of approximately 1.5 K between exposed and shadowed panels. The custom Stefan-Boltzmann boundary condition confirms that the system reaches a stable global thermodynamic equilibrium, continuously rejecting both the internal electronic heat and the absorbed solar energy into the 3.0 K deep space sink.
- **Design Insight:** The near-zero velocity stagnation zones identified on the inner radii of the 90° bends represent localized convective deficiencies. Future design iterations should avoid placing thermal mounting brackets at these locations and consider the use of rounded inner corners or guide vanes to mitigate flow separation and improve convective uniformity.

Future developments will extend this framework to transient orbital cycling analyses, evaluating the thermal response during alternating sunlit and eclipse phases, and exploring the effect of surface coatings with higher emissivity on the chassis radiative rejection capacity.

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Summary on Choice of Project and Methodology

Project selection and physical scenario. The thermal management of small satellites lies at a relevant intersection between space engineering, fluid mechanics, and multi-mode heat transfer — all core topics of the Master's programme. The CubeSat platform was selected because its high power-to-volume ratio and constrained surface area make it a challenging and well-bounded thermal design problem with direct industrial relevance in the rapidly growing commercial small satellite market. The decision to model the sunlit orbital phase was deliberate: it represents the most thermally demanding operational condition the satellite routinely experiences, combining a 1.0 W internal electronic load with a directional external solar influx of 408 $\mathrm{W/m^2}$, while the only available dissipation path is radiation to the 3.0 K deep space sink. Modeling a geometrically simplified but physically realistic geometry allowed the group to focus on the coupling between the three heat transfer modes without introducing unnecessary complexity.

Geometry simplifications. Omitting solar panels and modeling the entire electronics suite as a single equivalent solid heat source are deliberate simplifications. These reduce CAD and meshing complexity while preserving the physically relevant thermal pathways: conduction through the aluminum frame, convective absorption by the cooling loop, and radiative rejection from the outer surfaces. The perimeter cooling loop layout follows established engineering practice for active thermal control in CubeSat platforms [1]. The use of acrylic plates as directional thermal insulators was introduced to channel heat preferentially toward the active cooling path, demonstrating awareness of thermal insulation design in aerospace systems.

Radiation modeling. The use of custom Stefan-Boltzmann User Field Functions in lieu of the built-in Surface-to-Surface (S2S) or Discrete Ordinates (DOM) radiation models was a conscious trade-off. For a surface radiating to a uniform background temperature (deep space) with no inter-surface radiative exchange, the direct boundary condition is mathematically equivalent to the full radiation model at a fraction of the computational cost. Moreover, the explicit formulation allows independent control of emissivity and solar absorptivity on each face, enabling the asymmetric orbital environment to be modeled in a transparent and physically interpretable way. A recognized limitation of this approach is that it does not capture radiative exchange between satellite surfaces (inter-surface view factors); this would require a view-factor model in a more detailed follow-up study.

Pump modeling. The velocity-aligned volumetric momentum source approach (Eq. 4) avoids the significant added complexity of a rotating reference frame or sliding mesh. The velocity-normalized direction vector ensures the source always reinforces the local flow direction without introducing spurious transverse body-force components. This is standard practice in system-level cooling loop simulations and is proportionate to the required level of modeling detail for a steady-state analysis.

Mesh strategy. Prism Layers in the fluid domain were essential to resolve the thin thermal boundary layer at the pipe wall, where convective flux gradients are steepest and where the accuracy of the conjugate heat transfer coupling is most sensitive. The Thin Mesher for the pipe shell addressed the geometric challenge of its extremely small wall thickness. The Trimmed Cell Mesher for the structured solid components ensured orthogonal cell alignment

along the primary conduction paths. The mesh independence study, validated with both velocity and temperature convergence histories across three grids, provides quantitative evidence that the selected Medium mesh solution is free of spatial discretization error, with both monitored quantities varying by less than 1.25 % relative to the Fine mesh.

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